

SEE LION: Search and Emergency Equipment for Live-saving Identification and Optical Network

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I. Abstract

The Columbia Space Initiative has designed the Columbia SEE LION, an electronic system that utilizes image recognition techniques to identify life-saving objects of interest near the crash site of the Orion capsule in the Pacific Ocean. The system features a single-board computer to run the recognition model and process live video from a wired camera to identify all possible objects within the frame of the video, with results displayed onto an operator laptop via ethernet data transfer. Testing has confirmed that the SEE LION operates successfully in the Neutral Buoyancy Lab (NBL) pool-side testing environment, successfully identifying the life ring and life raft objects, and also highlighted areas of improvement for operator usability and functionality to address in future iterations of the SEE LION.

II. Introduction

As NASA rises to the challenge of advancing to the Moon with the Artemis program, crew safety and emergency preparedness remain paramount. When the Orion capsule returns to Earth, or in the case of an unexpected water landing after launch, debris from the capsule, such as safety gear, may be dispersed throughout the impact site, indicating the presence of crew members outside the capsule waiting to be rescued. To safely and efficiently handle search and rescue efforts to collect the astronauts with as much ease as possible, authorities should be equipped with enough information to quickly identify the whereabouts of the crew members at the capsule area of impact. Autonomous techniques can provide reliable, rapid assistance in informing authorities of potential crew locations in the water, efficiently rescuing all members of the Artemis crew.

The Columbia SEE LION, a Micro-g NExT search and rescue target recognition system designed by the Columbia Space Initiative, is a camera system that relies on the You Only Look Once (YOLO) image recognition model to identify objects of interest, such as orange-colored life rings and mannequins wearing life preserver units. Once the operator initializes the single-board computer via command terminal prompts, the image recognition model is deployed on the live camera video feed. Once an object of interest is detected, a data burst containing text with the object detected, and an image of the object with a bounding box and identified class is returned to the operator within 60 seconds to search suspected areas where astronauts can be found quickly.

III. Proposed Design

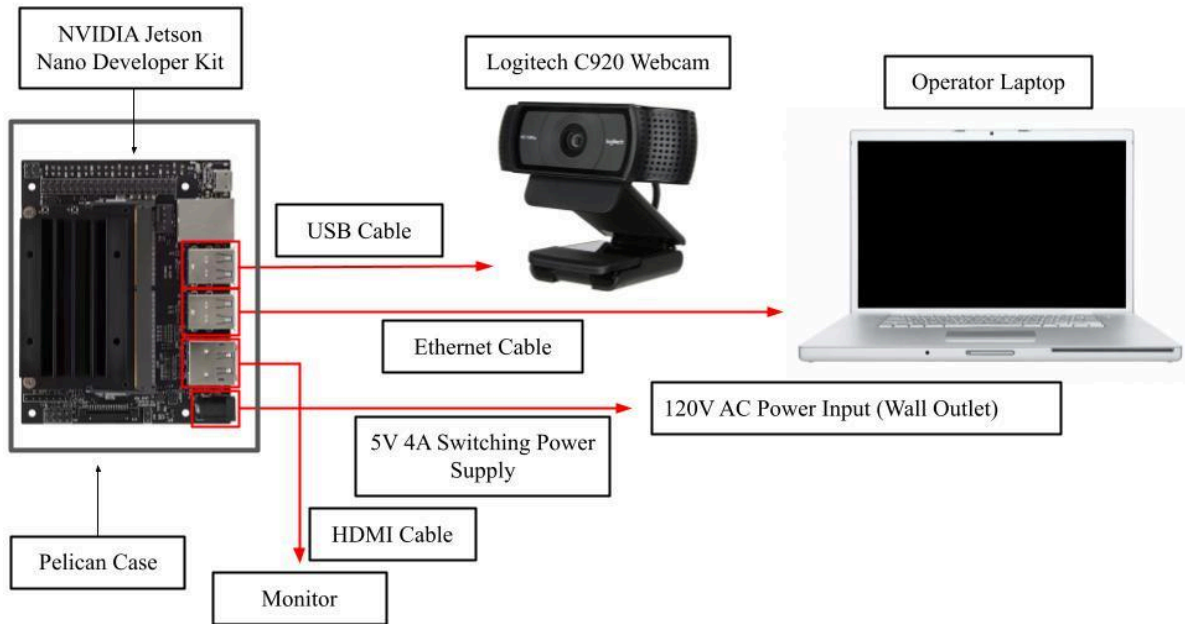


Figure 1. Labeled high-level diagram of the primary hardware for the SEE LION.



Figure 2. Labeled image of the SEE LION system with the user interface active on the laptop.

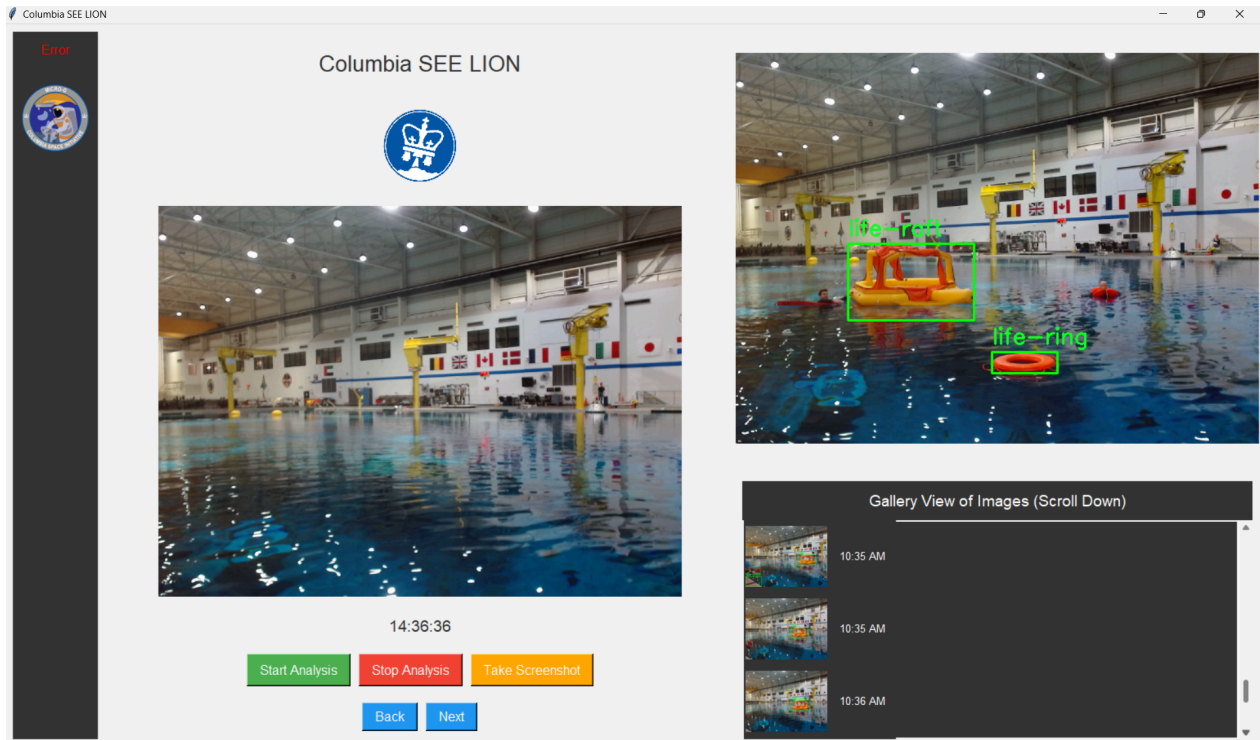


Figure 3. Screenshot of the SEE LION user interface displaying two identified objects at the NASA Neutral Buoyancy Laboratory. Official testing conducted June 5, 2024.

Dimensions

Weight:

Primary configuration: 6 lbs

Length, Width, and Height:

Pelican Case: 13.37 in. x 11.62 in. x 6 in.

Tripod with Camera (min. height/max. height): 25 in./52 in.

Overview

The SEE LION is an electronic system that relies on a pre-configured single-board computer with a custom-trained You-Only-Look-Once (YOLO) object recognition model to identify life-saving objects of interest from live video. The model is trained on images of the objects from the NBL to ensure consistent and accurate object identification, and the operator laptop user interface is designed for ease of use to minimize setup time during a search and rescue operation. With this in mind, a Logitech C920 Webcam provides the SEE LION with a live feed from its USB output, which is connected to an NVIDIA Jetson Nano Developer Kit. This will allow the Jetson Nano to detect objects and display alerts on the connected operator's laptop. Though there are various wireless options for connectivity between the camera and the Jetson Nano, as well as between the Jetson Nano and the operator laptop, a completely wired approach was selected with the SEE LION's robustness in mind, as a wireless connection might cause the SEE LION to be overly dependent on internet connectivity. This can reduce the system's functionality should there be issues due to a weak internet connection.

Camera

The Logitech C920 Webcam was selected due to its native compatibility with the Jetson Nano and its small size and weight for a more portable system design. The USB-A connection from the camera to the Jetson Nano allows for seamless video transfer, providing a robust option as Wi-Fi may not always be available when using the system.

Tripod System

The tripod system will be a splashproof aluminum alloy tripod system for poolside use. The maximum height of 50 inches was used for testing during setup and placement on the pool deck. The tripod comes with a mount to attach the Logitech webcam directly.

Single-Board Computer

The NVIDIA Jetson Nano Developer Kit was selected as the SEE LION's single-board computer to handle the computing required for the YOLO image recognition model. The Jetson Nano is connected to the camera via a USB-A connection and to the operator's laptop via an ethernet cable. Wiring is not a significant problem, with zip ties used to organize wiring inside the Pelican case.

Pelican Case

The Pelican 1400 Case was selected to house the Jetson Nano to ruggedize and waterproof the SEE LION system. This case protects the electronics using a custom enclosure panel installed on top of the case to access the ports on the Jetson Nano. The case also provides ample space for cables connecting to the Jetson Nano from the waterproof panel connectors.

Physical Device Setup

Although team members performed the physical setup before the testing session at the NBL, the following instructions are provided for users unfamiliar with the SEE LION with sufficient information to set up the device correctly.

- 1) Open the pelican case to find a power cable and a gray USB-to-Ethernet adapter. The top layer within the case should have five ports. From top to bottom and left to right, these ports are USB-A port, blue Ethernet cable, USB-A port, HDMI port, and circular power jack. Before beginning the operation of the SEE LION, users should familiarize themselves with this setup, identifying and setting aside the gray USB-C-to-Ethernet adapter and power cable with a circular plug.
- 2) Mount the separate camera to the silver tripod. Twist in place to tighten.
- 3) Set up the tripod by separating its legs, flipping out the clips on the tripod legs to allow it to telescope outwards, and flipping the clips to close when the tripod is at a satisfactory height.
- 4) Place the tripod so the camera is centered at the desired angle, facing the pool.
- 5) Plug the camera's USB-A cable into the USB-A port on the right side of the pelican case. The camera should light up in a few minutes.
- 6) Plug the gray USB-C-to-Ethernet adapter into the USB port of the operator's laptop.
- 7) Plug the Ethernet cable into the gray USB-C-to-Ethernet adapter. By this point, the SEE LION should be set up to run.
 - a) On a scale from 1 to 5, rate the ease of physically setting up the device (handling the Pelican case, connecting cables, turning on the laptop, running the user interface file).

IV. Device Testing

Testing Objectives

Substantial consideration was given to determining meaningful testing objectives for the SEE LION prior to the test session at the NBL. Although the device could be tested in an ordinary pool environment with near-equivalent objects of interest, the primary focus of the Test Week session was to test the system from both a functional and usability perspective. The team's primary concern was to ensure that the recognition model could identify each object of interest with a reasonably high degree of accuracy, at least above 50% for each object class. This metric was determined based on the object class confidence intervals of the model, as well as testing conducted with the model before the test session at the NBL. To quantify these results, a framework was developed to keep track of the number of objects deployed in the pool during the test, followed by objects successfully identified, objects not identified, and objects misidentified, noting both unidentified and misidentified object classes.

To quantify the usability of the SEE LION, the team asked users to rate many qualities of the device and their experience on a scale of 1 to 5, with 1 indicating that the feature did not function at all and 5 indicating that the feature functioned as required without flaw. Keeping a consistent scale was crucial for cross-analysis of the different components of the system. The team also asked users to recommend additional features that would make the current user interface setup more straightforward and convenient to use, as well as other helpful information to display that would be most useful to operators when deploying the SEE LION in a live search and rescue scenario. A sample

testing procedure is included below, showcasing the integration of the quantitative 1 to 5 scale and the detailed instructions to make the user's experience with the SEE LION as seamless as possible. Italics denote commands executed in the terminal.

Written Testing Procedure

- 1) Open the three command line terminals by pressing the "Win" key on the keyboard and typing "cmd." Repeat this step two more times.
- 2) In the first command line terminal, run the Docker to run the model offline. Run the following commands by pressing the up arrow until the terminal window displays the output below, then press "Enter":

First Terminal:

```
sudo docker volume create roboflow docker run -it --rm -p 9001:9001 --runtime=nvidia --mount source=roboflow,target=/tmp/cache roboflow/roboflow-inference-server-jetson-4.6.1
```

- 3) Wait approximately 15 to 20 seconds, then run the Python file containing the YOLO model in the second terminal. Run the following commands by pressing the up arrow until the terminal displays the following outputs, and press "Enter" for each line.

Press the up arrow until the terminal displays the following output:

```
cd roboflow-jetson-license-plate
```

Press "Enter".

Press the up arrow until the terminal displays the following output:

```
source plate2/bin/activate
```

Press "Enter".

Press the up arrow until the terminal displays the following output:

```
python3 infer-simple-new.py
```

- 4) Monitor the temperature of the Jetson Nano in the third terminal. The maximum operating temperature is 97 degrees Celsius or 97000, as shown on the terminal. Run the following command by pressing the up arrow until the output below is displayed.

```
cat /sys/class/thermal/thermal_zone0/temp
```
- 5) Expect to wait approximately 30 seconds to 2 minutes for the Jetson Nano to connect and the program to initialize. In the top left corner of the monitor, there should be two boxes displaying "connecting" or "initializing."
- 6) The SEE LION should be able to operate for approximately 5 to 7 minutes while running the inference server before overheating issues arise. While the SEE LION is in operation, ask the following questions:
 - b) On a scale from 1 to 5, rate the ease of engaging with the user interface. How well were you able to navigate the functionality and the features provided?
 - c) On a scale from 1 to 5, rate the ease of finding the objects of interest from viewing the user interface.
 - d) What additional features would make the current user interface setup more straightforward and convenient to use?
 - e) What additional information would be most beneficial to you as an operator when deploying the device in a real search-and-rescue scenario?

Results

During the NBL test, results were successfully collected using the abovementioned procedures after explaining the steps for device usage and the above rating scale. Though there were no setup issues, the user did experience operation issues due to software, as the integrated development environment kernel needed to be restarted. Of the four target items deployed, the SEE LION correctly identified two objects, with no misidentified objects. As such,

the SEE LION met the team's goal of ensuring that the recognition model could identify the objects of interest with an accuracy of at least 50%. However, two classes of target items were also not identified: the orange-colored life preserver unit and the Orion capsule mockup on the pool's far end. This suggests that additional data for the unidentified object classes may be required.

The feedback on the running of the SEE LION noted the video output displayed a low framerate. A buffer was recommended to improve the lag and reduce the time required to initialize the program. To keep human operators in the loop of the SEE LION's operations by allowing them to observe the margin of error, the operators also recommended adding confidence intervals to the SEE LION's detections. However, they noted that this was beyond the scope of the challenge description. Based on their feedback, additional information that would be most useful to operators deploying the device in a real search-and-rescue scenario included data on the distance and direction of the object detected and better camera resolution.

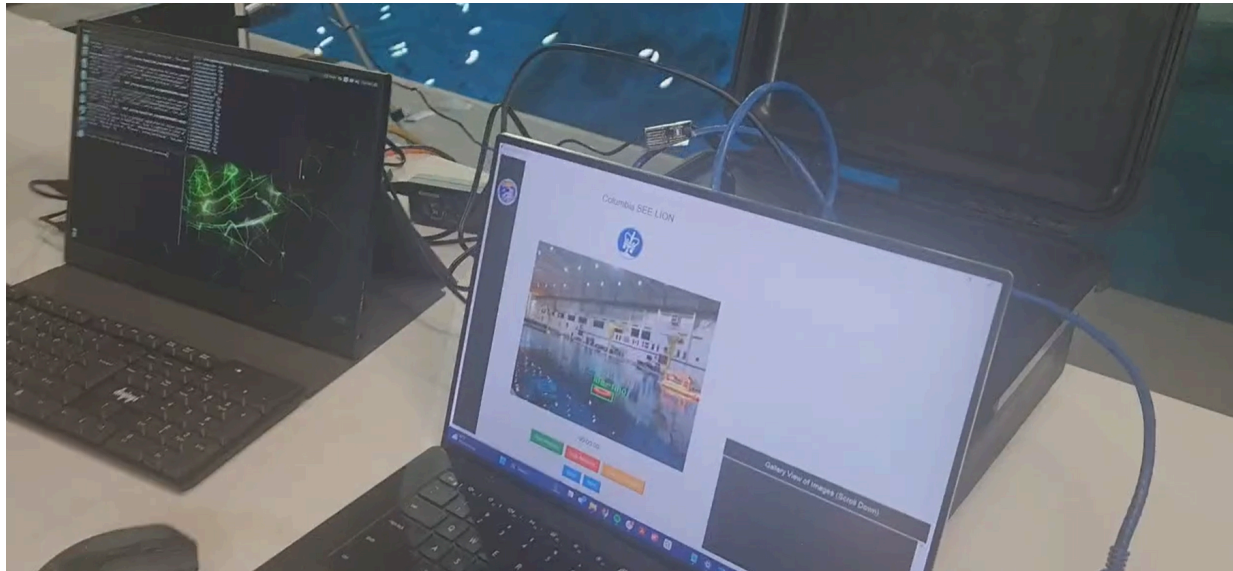


Figure 4. Video capture of the poolside test setup at the Neutral Buoyancy Laboratory.

V. Future Iterations

In future iterations, the collection of additional accurate and realistic data, namely images and video capture of the object classes, will be prioritized to improve the precision and recall of the YOLO model. Before testing at the NBL, our model was trained on approximately 200 images, collected by finding images online and generating synthetic data with the photos provided by the Micro-g NEXt team. Collecting data on objects at the NBL before the testing session enabled the system to detect the deployed objects within a familiar environment based on the newly collected training data.

To improve the usability of the SEE LION, task management software like CronJob will be used instead of bash scripts for events to trigger actions to execute multiple tasks for the program, streamlining the setup of the SEE LION software script. To reduce the lag for a smoother run of the SEE LION, a buffer will be introduced to prevent wait times for the program to initialize. To deliver beyond the scope of the challenge description, with additional time and resources, further information that would be most useful to operators deploying the device in a real search-and-rescue scenario would be added, including confidence intervals to keep human operators in the loop of the SEE LION's operations by allowing them to observe the margin of error, data on the distance and direction of the object detected, as well as better camera resolution.

Prompted by questions from the team, the user gave detailed feedback at each stage of the testing process. Beginning with the setup of the SEE LION's physical connections, though the operators did not directly set up the SEE LION but were instead given a detailed explanation of the setup while watching the program being launched, the operators were still able to recommend improvements to streamline the setup of the SEE LION software script.

For example, to execute multiple tasks for a program, the operators recommended task management software like CronJob instead of bash scripts for events to trigger actions, which will help mitigate issues related to connectivity in both ethernet and wireless connection.

VI. Conclusion

Our team designed the SEE LION system to enable search and rescue respondents to focus on what matters most when it comes to the critical and time-sensitive duty of helping rescue astronauts from the Orion capsule upon impact in the Pacific Ocean, using cutting-edge technologies such as the YOLOv7 object detection framework, and small but powerful hardware in the Jetson Nano, to provide the computing power required for the task.

During our time at the Neutral Buoyancy Laboratory, we gathered valuable feedback from the challenge owners that can be used to build upon the device's current functionality and usability. We found that while our device could identify objects of interest with a reasonably high degree of accuracy at 50%, there are several opportunities to improve the SEE LION's ability to identify all four object classes in the future. Furthermore, the user experience for the operator utilizing the SEE LION can be improved to provide a streamlined process for using the device, which can save critical time when deploying the device in a real search and rescue environment.

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